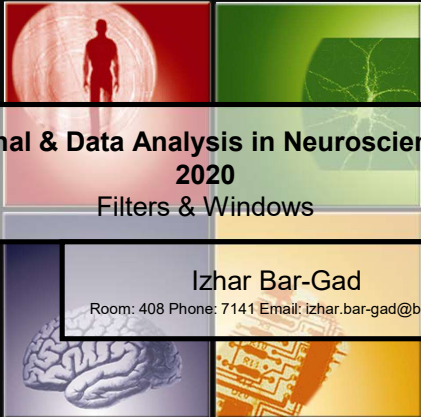



Signal & Data Analysis in Neuroscience 2020

Filters & Windows

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Outline – Frequency Domain

- ☑ Introduction
- ☑ Fourier Transform
- ☑ Sampling Theory
- ☑ Systems
- **Filters**
- Windows
- Spectral Analysis

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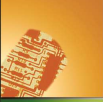
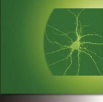
Filter in signal analysis

- Filters are used to **eliminate** unwanted frequencies from an input signal or to **select** a desired frequency range among many others.
- Filters form a sub-group of the systems.
- There are analog and digital filters. We will focus on the digital filters.

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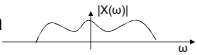


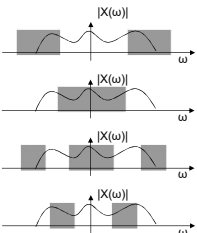
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Filters

Original frequency domain

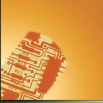


- Low Pass Filter (LPF)
- High Pass Filter (HPF)
- Band Pass Filter (BPF)
- Band Stop Filter (BSF)



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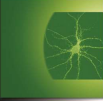

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Input – Filter – Output

- Given the input $X(\omega)$ and desired output $Y(\omega)$, how can we describe $H(\omega)$?
- Moreover, how can we transfer it to the time domain and get a representation of the three components?

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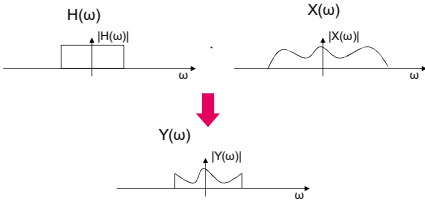


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Ideal Filtering

Since $Y(\omega) = X(\omega) \cdot H(\omega)$



Remember : No frequency domain systems exist !!

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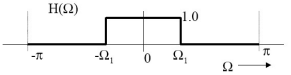





Ideals problem : $F(\text{rect})=\text{sinc}$

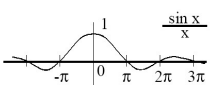
Ideal frequency response $\rightarrow$ sinc in the time domain !

example:
ideal low-pass



$$h[n] = \frac{1}{2\pi} \int_{-\Omega_1}^{\Omega_1} \exp(j\Omega n) d\Omega = \frac{1}{2\pi} \left[\frac{\exp(j\Omega n)}{jn} \right]_{-\Omega_1}^{\Omega_1}$$

$$= \frac{1}{2\pi jn} [\exp(j\Omega_1 n) - \exp(-j\Omega_1 n)]$$

$$= \frac{1}{n\pi} \sin(n\Omega_1) = \frac{\Omega_1}{\pi} \text{sinc}(n\Omega_1)$$


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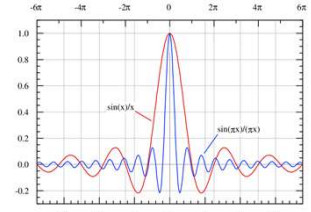
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The sinc function

- The classical "unnormalized" definition $\text{sinc}(x) = \frac{\sin(x)}{x}$
- Signal analysis "normalized" definition $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$



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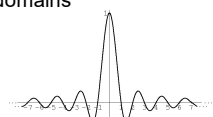



sinc problem

- The sinc function is infinite, forever oscillating around zero.

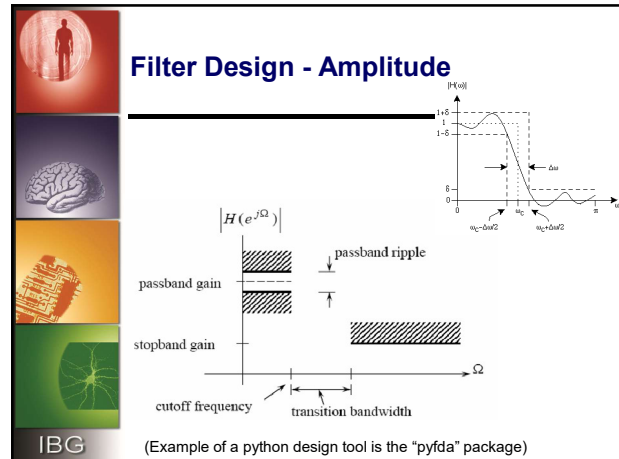
↓

- Inserting Irrelevant samples into the filtered signal
- Since both $F(\text{rect})=\text{sinc}$ & $F(\text{sinc})=\text{rect}$ this problem is relevant for rectangular windows in both time and frequency domains

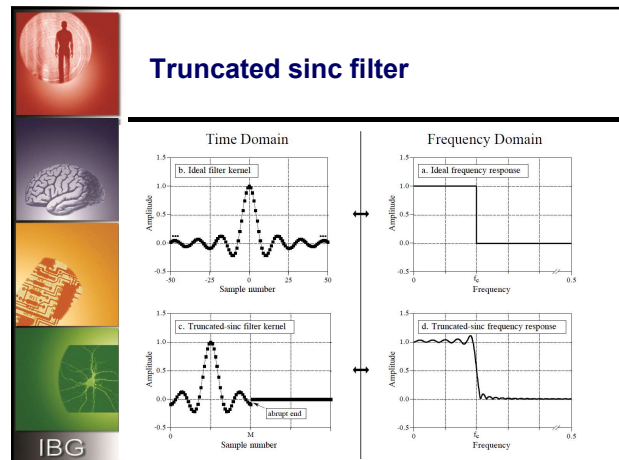


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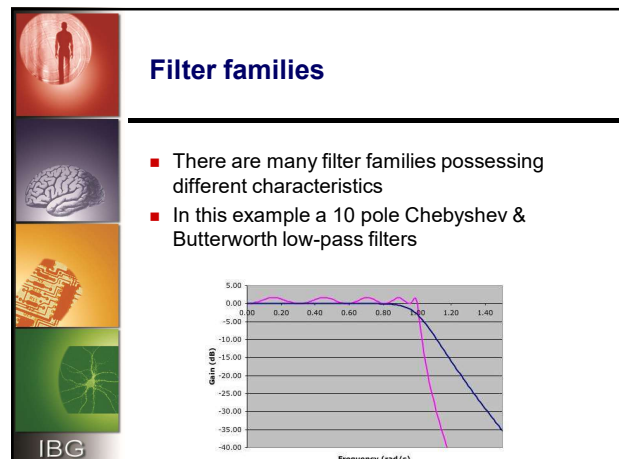
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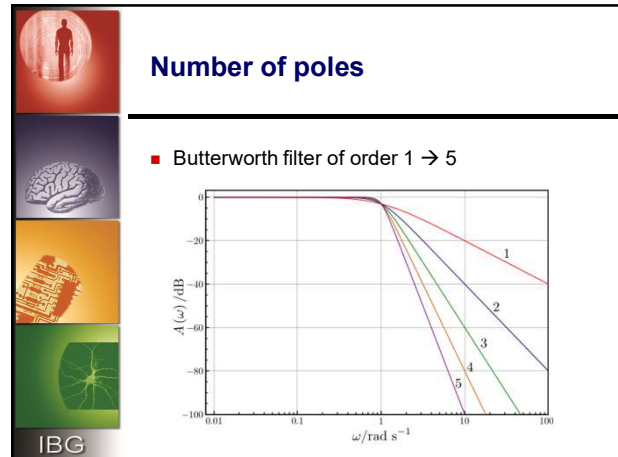
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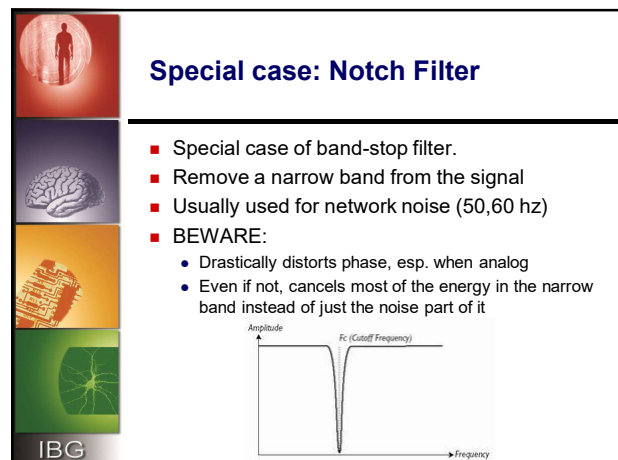
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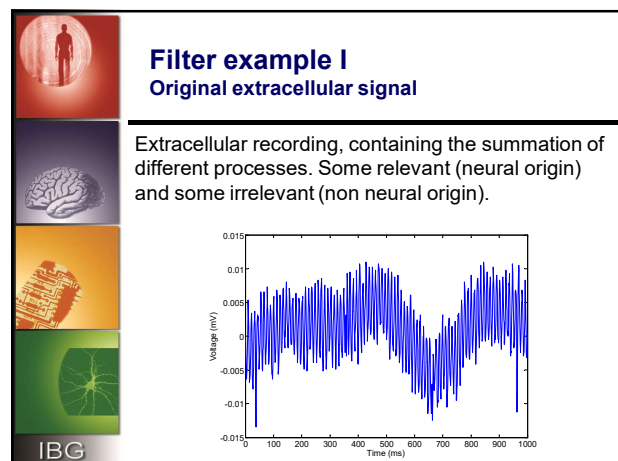
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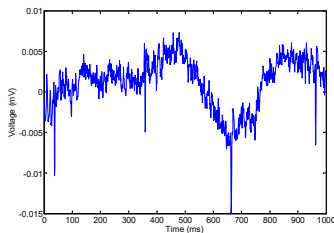





Filter example II

Following notch filter (~1800Hz)

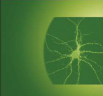
External (non biological noise) may have a very narrow band. Both local field potential (LFP) and action potentials (spikes) are apparent.



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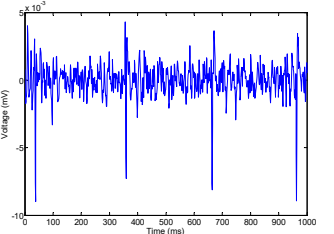


Filter example III

Following high pass (~300Hz)

Removal of the high frequency components removes the local field potential, leaving the spiking activity.



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Filter Design - Phase

LTI system: $H(\omega) = A(\omega)e^{j\theta(\omega)}$

↓

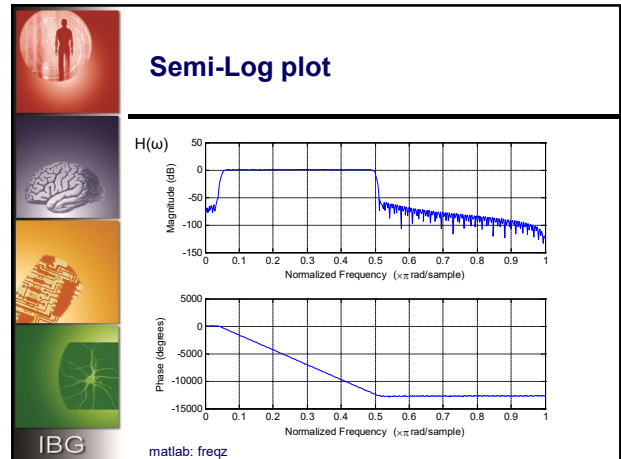
$$P(\omega) \triangleq -\frac{\Theta(\omega)}{\omega}$$
(Phase Delay)

Therefore, in order to achieve **constant delay** (that all frequencies will be delayed by the same phase) we would like to have a **linear phase response**.

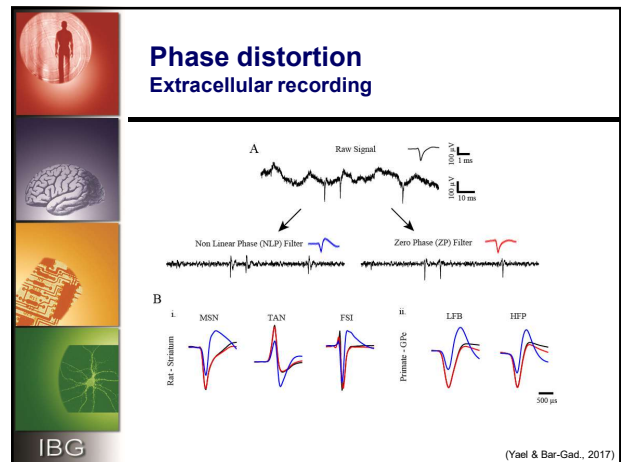
[or at least at the relevant frequency range]

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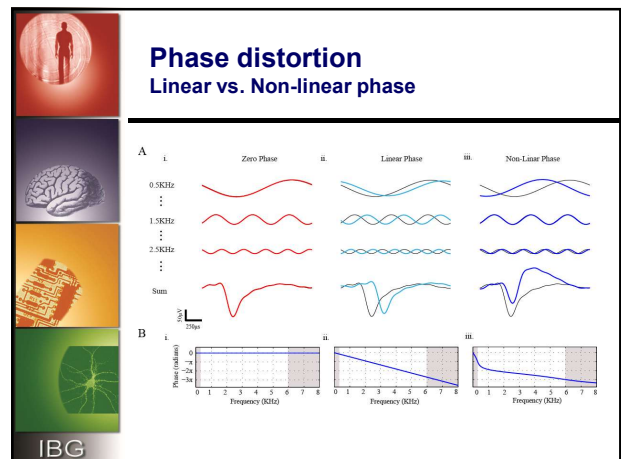
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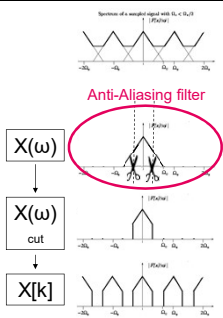
Filters

Avoiding sampling problems

when $X(\omega) \rightarrow X[k]$ looks like this,
You have an aliasing problem...

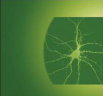
What is there to do?

And if we can not increase ω_s ?



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Online vs. Offline Filtering

- Online filtering –
 - Live streaming (usually analog) data
 - Delay limit
 - Limited computational resources
- Offline filtering –
 - Sampled, stored, digital data
 - No time limit
 - Less limited computational resources

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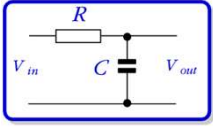
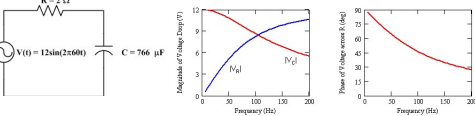





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Analog Filters

RC Circuit = the simplest LPF

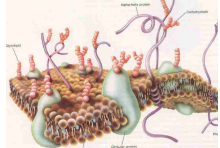
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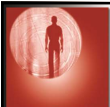



Filters in Neuroscience

- Cell Membrane as LPF
- Slice Bath as LPF

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2D Fourier transform

- Continuous 2D transform

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

- Discrete 2D transform

$$F[k, l] = \frac{1}{\sqrt{MN}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f[m, n] e^{-j2\pi(\frac{mk}{N} + \frac{nl}{M})}$$

$$f[m, n] = \frac{1}{\sqrt{MN}} \sum_{l=0}^{N-1} \sum_{k=0}^{M-1} F[k, l] e^{j2\pi(\frac{mk}{N} + \frac{nl}{M})}$$

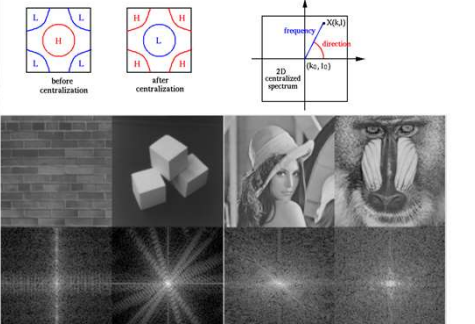
(http://fourier.eng.hmc.edu/e101/lectures/Image_Processing/Image_Processing.html)

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2D Fourier transform



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Spatial filtering

- Spatial filters may be applied on 2D (or even 3D) images.

<https://medium.com/@hrcraigchen/digital-image-processing-using-fourier-transform-in-python-bcb49424682>

Original

Low pass

High pass

(from OpenCV github)

Spatial filtering

Ideal filter

Butterworth filter

Low pass

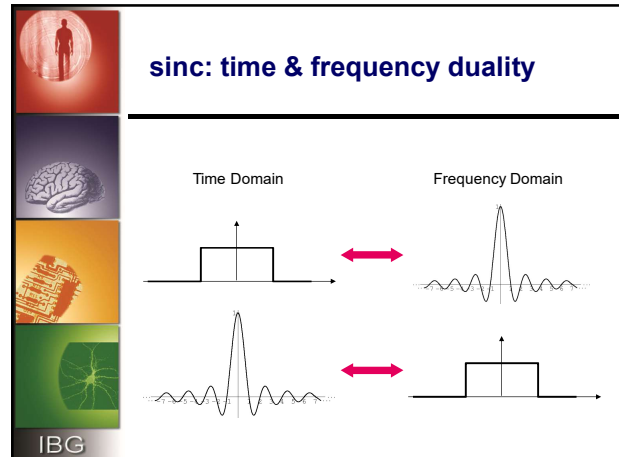
High pass

(from OpenCV github)

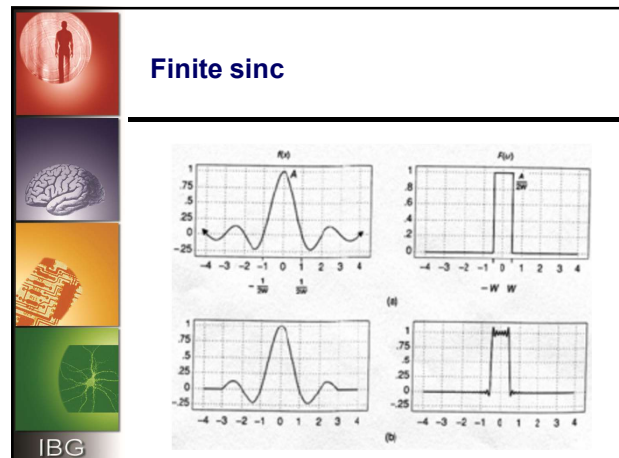
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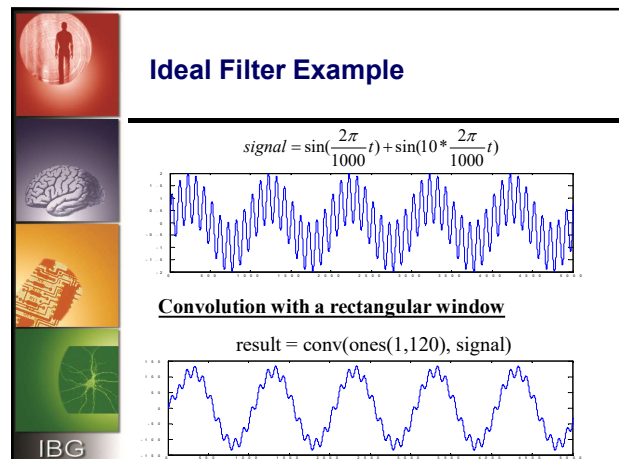
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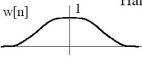



Windows

rectangular window - sharp edges - spreads out in frequency domain
we seek a more gentle window function, tapered towards the edges

$$w[n] = 0.5 + 0.5 \cos\left(\frac{n\pi}{M+1}\right)$$

$$-M \leq n \leq M$$



“Hanning”

$$w[n] = 0.54 + 0.46 \cos\left(\frac{n\pi}{M}\right)$$

$$-M \leq n \leq M$$



Hamming








Windows frequency response

frequency

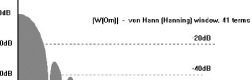


Triangular window

$$W[n] = 1 - \frac{2}{M-1} \left| n - \frac{M-1}{2} \right|$$

$$-M \leq n \leq M$$

Hanning window

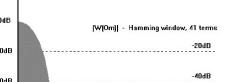


van Harn [Hanning] window, 41 terms

$$w[n] = 0.5 + 0.5 \cos\left(\frac{n\pi}{M+1}\right)$$

$$-M \leq n \leq M$$

Hamming window



Hamming window, 41 terms

$$w[n] = 0.54 + 0.46 \cos\left(\frac{n\pi}{M}\right)$$

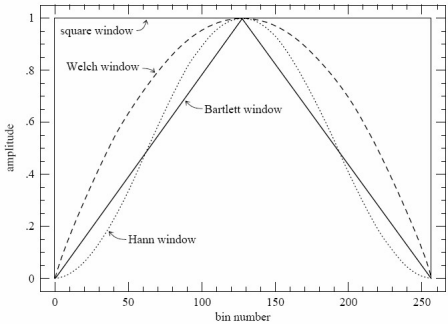
$$-M \leq n \leq M$$



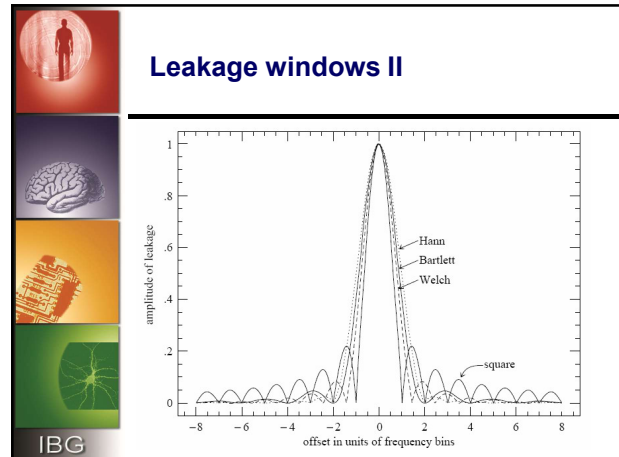




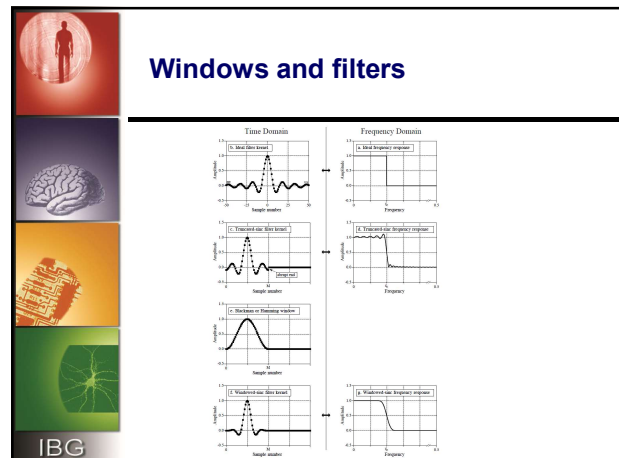

Leakage windows I







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DFT Frequencies

- Discrete
- Range : $0-2\pi$, $e^{-j\frac{2\pi k}{N}n} \rightarrow \Delta k = \frac{2\pi}{N}$
- Periodic

But...

What about frequencies in between?

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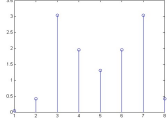


DFT Leakage

Say we had a sinusoid of 2.3 KHz and we still sampled it at 8000 samples/s :

The output frequency bins = 0KHz, 1KHz, ..., 7KHz
2.3 KHz bin is missing!!

What happens is that this frequency component shows up (leaks) in all the other frequency bins



Remedy
Windowing Techniques